# Transistors

Fundamental Building Block of Modern Electronic Circuits

Transistor defined: *a semiconductor device controlling current flow to amplify or switch signals*.

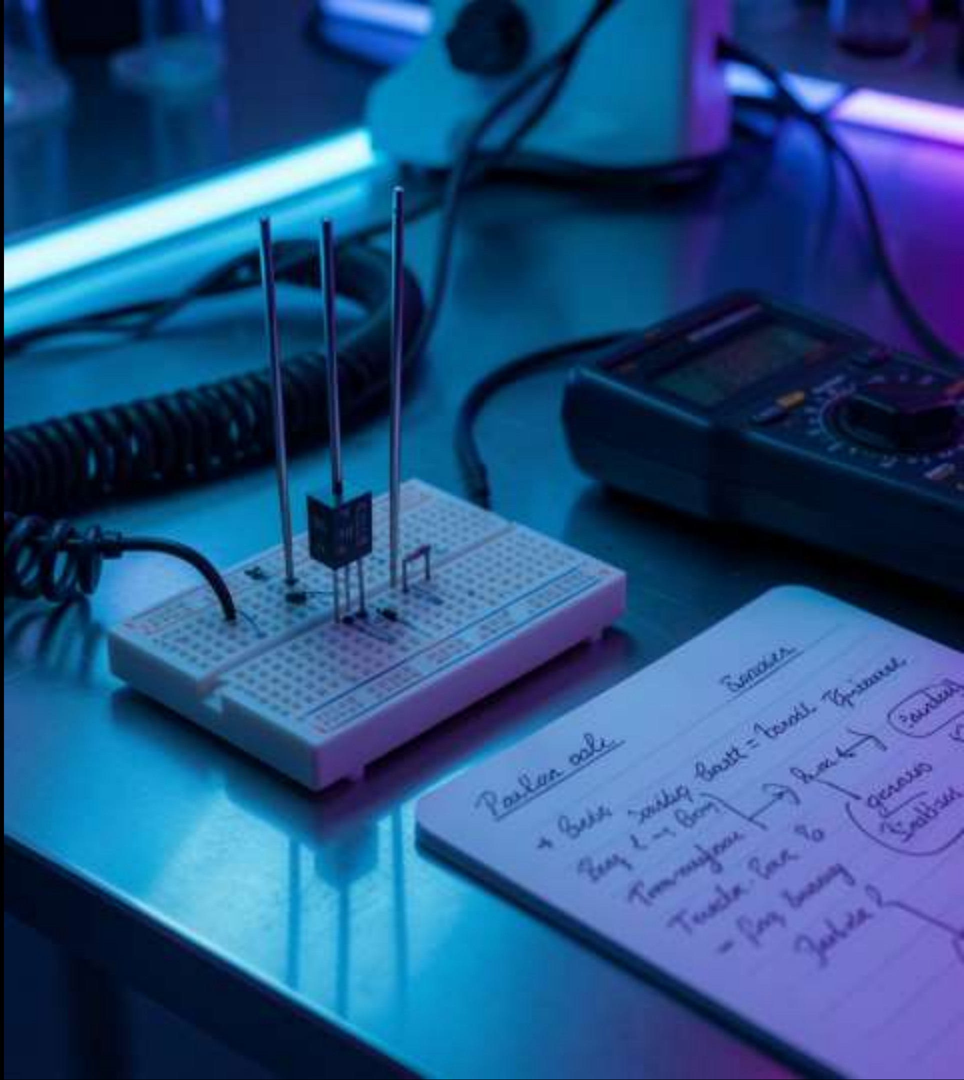
Brief historical note: Invented in 1947; cornerstone of modern electronics.

Speaker notes: Emphasize invention year and role as a replacement for vacuum tubes. Mention diagram placeholder for transistor symbol and basic circuit.

# Introduction

# 01

# Transistor Basics

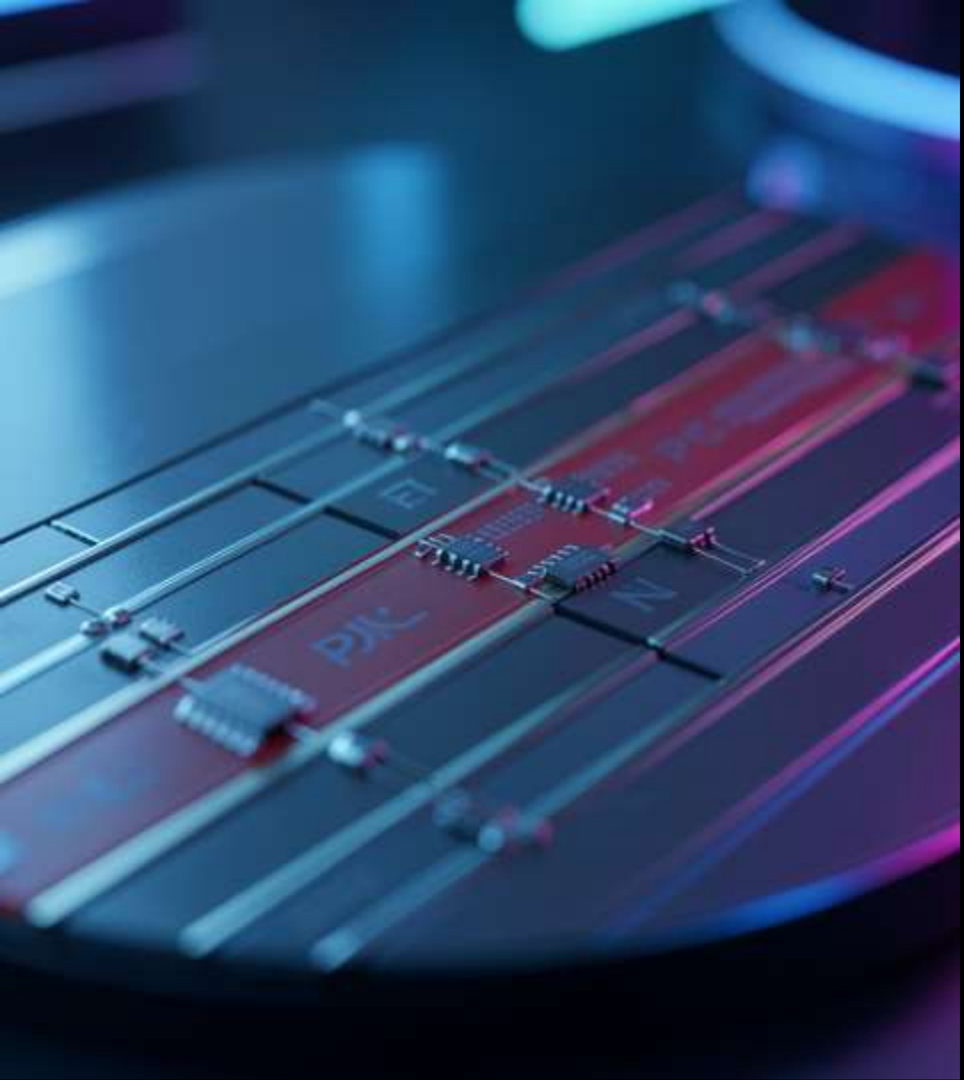


# Definition & History

*Definition:* Solid-state device controlling current between terminals to **amplify** or **switch**.

History: Invented 1947 at Bell Labs; enabled miniaturization and modern computing.

Speaker notes: Highlight impact on electronics evolution and transition from vacuum tubes. Show timeline slide graphic and transistor symbol diagram.



# Structure & Materials

Materials: Silicon and Germanium; doped to form *N-type* and *P-type* regions.

Structure: PN junctions form BJT and channel/oxide form FET/MOSFET.

Diagram: transistor structure and circuit symbols (placeholders).

Speaker notes: Explain doping, PN junction role, and differences between BJT and MOSFET structures. Point to diagrams for clarity.

# PN Junctions

A PN junction forms where *P-type* and *N-type* semiconductors meet, creating a depletion region that controls carrier flow.

Forward bias reduces the barrier enabling current; reverse bias widens barrier blocking current.

**Key takeaway:** PN junctions are the fundamental switching and rectifying element in transistors.

# 02

# Applications & Types

# Types: BJT / FET / MOSFET

BJT: Current-controlled, uses *PNP/NPN* structures for amplification and switching.

FET: Voltage-controlled, high input impedance; MOSFET adds oxide layer for insulation.

**Key takeaway:** Choose BJT for drive, MOSFET for efficiency and scaling.

# Configurations & Operation

Common Emitter: High gain, phase inversion — used in amplifiers.

Common Base: Low input impedance, high frequency stability.

Common Collector (Emitter Follower): Voltage buffering, low output impedance.

**Key takeaway:** Configuration determines gain, impedance, and phase behavior.

Uses: Signal amplification, digital logic switches, power regulation, RF communication.

Advantages: **small size**, low power consumption, high integration density.

Limitations: Thermal sensitivity, breakdown voltages, and scaling challenges at nanoscale.

# Uses & Limitations

# Conclusions

Transistors are compact semiconductor devices that *amplify* and *switch* signals, enabling modern electronics.

Their diverse types and configurations suit applications from microprocessors to power systems.

**Final point:** Continued scaling and material innovation remain essential for future performance gains.

# THANKS!

Do you have any questions?

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